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# Aliasing Artifacts using Sampling and Quantization

import cv2
import numpy as np
import matplotlib.pyplot as plt
from google.colab import files

# Upload an image
uploaded = files.upload()

# Read the uploaded image
image_name = list(uploaded.keys())[0]
img = cv2.imread(image_name)
img = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)

# Create aliasing by excessive downsampling
small = cv2.resize(img, (img.shape[1]//8, img.shape[0]//8),
                    interpolation=cv2.INTER_NEAREST)

aliased = cv2.resize(small, (img.shape[1], img.shape[0]),
                     interpolation=cv2.INTER_NEAREST)

# Correct using bicubic interpolation
corrected = cv2.resize(small, (img.shape[1], img.shape[0]),
                       interpolation=cv2.INTER_CUBIC)

# Display images
plt.figure(figsize=(15,5))

plt.subplot(1,3,1)
plt.imshow(img)
plt.title("Original Image")
plt.axis("off")

plt.subplot(1,3,2)
plt.imshow(aliased)
plt.title("Aliased Image")
plt.axis("off")

plt.subplot(1,3,3)
plt.imshow(corrected)
plt.title("Corrected Image")
plt.axis("off")

plt.show()
```

Choose Files BIRD.webp

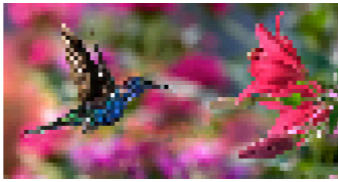
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Saving BIRD.webp to BIRD.webp

Original Image



Aliased Image



Corrected Image

